

Feed-Forward Active Noise Control System Using Microphone Array

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Abstract—Feedforward active noise control (ANC) system are widely used to reduce the wide-band noise in different application. In feedforward ANC systems, when the noise source is unknown, the misplacement of the reference microphone may violate the causality constraint. We present a performance analysis of the feedforward ANC system under a noncausal condition. The ANC system performance degrades when the degree of noncausality increases. This research applies the microphone array technique to feedforward ANC systems to solve the unknown noise source problem. The generalized cross-correlation (GCC) and steering response power (SRP) methods based on microphone array are used to estimate the noise source location. Then, the ANC system selects the proper reference microphone for a noise control algorithm. The simulation and experiment results show that the SRP method can estimate the noise source direction with 84% accuracy. The proposed microphone array integrated ANC system can dramatically improve the system performance.

Index Terms—Active noise control, causality, direction estimation, generalized cross-correlation (GCC), microphone-array, noise pollution, noise source, steering response power (SRP).

I. INTRODUCTION

NOISE pollution becomes a pressing problem due to the development of industrial applications [1], [2]. Reducing noise continues to be a challenge to maintaining and increasing the quality of life. Feedforward active noise control (ANC) systems are based on adaptive system identification, thus, are able to control both broadband and narrowband noises. Therefore, ANC systems are widely used in many practical applications, such as heating, ventilation, and air-conditioning systems [3], [4]; engine exhaust systems [5], [6]; and ANC headphones [7], [8]. However, when the acoustic/electric delays in the ANC systems exceed the acoustic delay of the primary path, the causality constraint will be violated [9]. The performance of the feedforward ANC system dramatically degrades as the degree of noncausality increases. Thus, the positions of the noise source and the reference microphone are critical for feedforward ANC system's performance.

In some ANC applications, the noise source position is known in advance, for example, electronic mufflers. Therefore, the reference microphone can be placed at the proper upstream

position to make the ANC filter casual. However, in many other applications, the noise source is unknown or moving, such as the ANC system for infant incubators [10]. The ANC system may work with a degraded performance or is unable to cancel the primary noise, as shown in Section IV. Therefore, there is an increased demand for ANC systems to estimate the noise source location or direction, then one can select the reference signal to cancel unwanted noise from unknown noise source or moving noise source environment, for example, a noisy street.

Microphone array is widely used in speech signal processing for speaker direction detection [11]–[15]. However, the noise sound signal, especially broadband noise, is highly uncorrelated which is different from speech signal. Therefore, the conventional localization methods designed for speech signal are difficult to utilize in noise signal situations.

In this paper, we propose to integrate a microphone array technique with the feedforward ANC system. The microphone array is used to estimate the noise source location or direction by using generalized cross-correlation (GCC) and steering response power (SRP) methods [16], [17]. The ANC system selects the microphone close to the noise source as the reference microphone, using the filtered-X least mean square (FXLMS) algorithm to reduce the unwanted noise level.

The paper is organized as follows, Section II presents the causality problem of the feedforward ANC system and analyzes the system performance. Section III proposes the microphone array integrated ANC system and noise source localization algorithm. Section IV shows the simulation and experiment results, and Section V concludes the paper.

II. PROBLEM STATEMENT

In the feedforward ANC system configuration illustrated in Fig. 1, the primary noise is sensed by a reference microphone. The anti-noise, which is the output of the adaptive filter, is played by the secondary loudspeaker and it passes through the acoustic path, then reaches the error sensor. The acoustic delay AD_1 from the reference microphone to the error sensor is proportional to the distance from the reference sensor to the error sensor. AD_2 represents another delay between the secondary loudspeaker and the error sensor. Since the adaptive filter necessarily has a causal response, we must ensure that the acoustic delay between the reference and the error microphones is greater than the electric delay ED , plus the acoustic delay from the secondary loudspeaker [9]. That is: $AD_1 > AD_2 + ED$, and this condition is called the causality constraint. When this constraint is violated, the response of

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the ANC system is noncausal, and hence not realizable for broadband noise control.

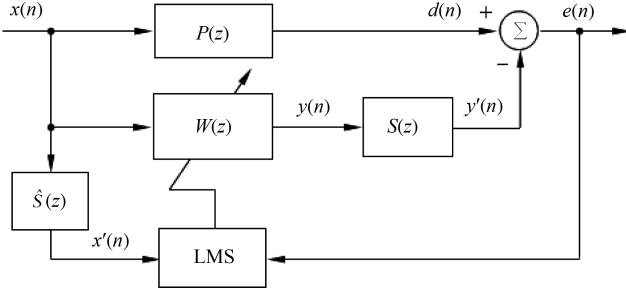


Fig. 1. The block diagram of a single channel feedforward ANC system.

Fig. 1 shows the block diagram of a single channel feedforward ANC system. The primary path $P(z)$ models the propagation path between the reference sensor and error sensor (including AD_1), while $S(z)$ is the secondary path between the anti-noise loudspeaker and the error sensor (including a part of ED and AD_2). $\hat{S}(z)$ is the estimation of the secondary path's transfer function. In order to analyze the causality effect, we simplify $P(z)$ and $S(z)$ as pure delays with their impulse response functions, $p(n) = \delta(n - \Delta p)$ and $s(n) = \delta(n - \Delta s)$. When $\tau = \Delta p - \Delta s > 0$, the system is causal. The Z domain signal can be expressed as

$$\begin{aligned} E(z) &= X(z)z^{-\Delta p} - X(z)W(z)z^{-\Delta s} \\ &= X(z)z^{-\Delta s}(z^{-\tau} - W(z)) \end{aligned} \quad (1)$$

and when $\tau = \Delta p - \Delta s < 0$, the causality constraint is violated, therefore we have

$$\begin{aligned} E(z) &= X(z)z^{-\Delta p} - X(z)W(z)z^{-\Delta s} \\ &= X(z)z^{-\Delta s}(1 - W(z)z^{-\tau}). \end{aligned}$$

In this case, adaptive filter $W(z)$ works as a predictor to minimize the residual error: $W(z) = z^\tau$.

The primary broadband noise can be modeled as an autoregressive (AR) model [18], [19]

$$x(n) = \sum_{i=1}^L a_i x(n-i) + n(n) \quad (2)$$

where $n(n)$ is a white noise with 0 mean and variance σ^2 , and α_i represents the constant coefficients for the AR model.

We assume that $\Delta p = 0$ without loss of generality, so the primary noise at the error sensor becomes

$$\begin{aligned} d(n) &= x(n) = \sum_{i=1}^L a_i x(n-i) + n(n) \\ &= \sum_{k=0}^{\infty} w(k)n(n-k) \end{aligned} \quad (3)$$

with $w(k)$ as the desired coefficient for this moving average (MA) model [20]. Meanwhile, the anti-noise generated from the adaptive filter $W(z)$ is

$$y(n) = \sum_{k=0}^{\infty} \tilde{w}_n(k)n(n-k-\tau) \quad (4)$$

where $\tilde{w}_n(k)$ is the estimated adaptive filter coefficients. The error picked at the error sensor is then $e(n) = d(n) - y(n) = \sum_{k=0}^{\infty} w(k)n(n-k) - \sum_{k=0}^{\infty} \tilde{w}_n(k)n(n-k-\tau)$. So the optimal $\tilde{w}_{opt}(k)$ is $\tilde{w}_{opt}(k) = w(n+\tau)$. The mean square error (MSE) can be achieved as

$$\begin{aligned} E[e^2(n)] &= \left[\left(\sum_{k=0}^{\infty} w(k)n(n-k) - \sum_{k=0}^{\infty} \tilde{w}_n(k)n(n-k-\tau) \right)^2 \right] \\ &= \sigma^2 \sum_{k=0}^{\tau-1} w^2(k). \end{aligned} \quad (5)$$

Therefore, the MSE is related with τ and the desired model coefficients $w(k)$. For the same model $w(k)$, MSE will increase while the degree of causality is increased.

There are many factors related to causality, and the most critical one is to select the reference sensor that is close to the noise source.

III. MICROPHONE ARRAY INTEGRATED ANC SYSTEM

In this section, we use an infant incubator ANC application as an example. We combine the microphone array technique with a feedforward ANC system. For an ANC application with unknown noise source/sources, The microphone array is used to estimate the noise sources' direction or location, then the proper reference sensor is selected and the ANC algorithm is conducted.

The ANC system contains K secondary sources and M error sensors, as shown in Fig. 2. The microphone array contains J ($J = 4$) microphones, and $m_1 - m_4$ are deployed at the edge or corners of the system. The noise source location or direction is estimated by the microphone array. The reference microphone is selected based on the noise source position for conducting the FXLMS algorithm.

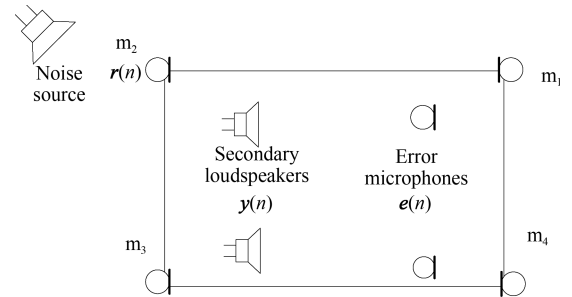


Fig. 2. Integrated ANC system ($m_1 - m_4$ are the sensors in the microphone array).

The primary noise is sensed by a reference microphone (m_2) close to the noise source [1, 2]. The adaptive filters use the sensed reference signal $\mathbf{r}(n) = [r(n) \ \dots \ r(n-L+1)]^T$ to generate the cancelling signal $\mathbf{y}(n) = [y_1(n) \ \dots \ y_k(n)]^T$ with K channels; which is fed to secondary loudspeakers to cancel the primary noise. The M error microphones measure the residual noise $\mathbf{e}(n) = [e_1(n) \ \dots \ e_m(n)]^T$ and use them to update the filter coefficients.

The noise source direction or position can be estimated by processing the signals received by the microphone array, as shown in Fig. 3.

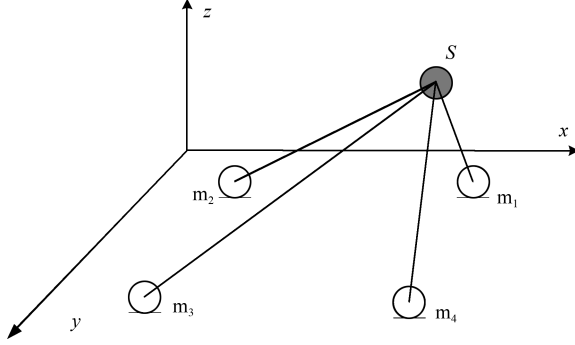


Fig. 3. Microphone array with 4 sensors.

Assume that S is a noise source with unknown coordinate (x, y, z) , the noise signal is picked up by microphones: $m_1, \dots, m_4$ simultaneously. Assume m_1 is at the coordinate system origin and the position of the i th microphone in the array is (x_i, y_i, z_i) with $i = 2, 3$ and 4 . Based on the geometry of the microphone array, the coherence and time difference among multiple copies of the same signal can be used to estimate the noise direction or location.

A. Noise Source Direction Detection

The time difference of arrival (TDOA) method is widely used in acoustic event localization. The TDOA estimation problem is to measure the time difference between the signals received at different microphones [11], [12].

Consider an unknown noise source S at (x, y, z) , and the sound travel time between S and i th microphone is

$$T_i = \frac{1}{v_s} \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} \quad (6)$$

where v_s is the speed of sound, and $i = 1, 2, 3, 4$.

Then, the TDOA between m_i and m_1 can be expressed as

$$\begin{aligned} \tau_{i1} &= T_i - T_1 \\ &= \frac{1}{v_s} \left(\sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} \right. \\ &\quad \left. - \sqrt{x^2 + y^2 + z^2} \right), \quad i = 2, 3, 4. \end{aligned} \quad (7)$$

Equation set (7) contains three hyperboloid equations with three unknowns (x, y, z) . By solving (7), we can obtain the position of the noise source.

1) *Generalized Cross-Correlation*: The generalized cross-correlation (GCC) algorithm is one of the most popular methods for TDOA estimation [17], which is defined as the expectation of two observed signals

$$\begin{aligned} r_{m_i m_j}^{\text{GCC}}(k) &= F^{(-1)}[\Psi_{m_i m_j}(f)] \\ &= \int_{-\infty}^{\infty} \Psi_{m_i m_j}(f) e^{j2\pi f k} df \\ &= \int_{-\infty}^{\infty} v(f) \Phi_{m_i m_j}(f) e^{j2\pi f k} df \end{aligned} \quad (8)$$

where $F^{-1}[\cdot]$ is the inverse discrete-time Fourier transform, $\Psi_{m_i m_j}(f)$ is the cross spectrum density, $v(f)$ is a frequency-domain weighting function, and $\Phi_{m_i m_j}(f)$ is cross-spectrum for two observed signals as

$$\Phi_{m_i m_j}(f) = E[M_i(f)M_j^*(f)]. \quad (9)$$

The maximum possible delay will be given where $r_{m_i m_j}^{\text{GCC}}(k)$ achieves its maximum at $\tau = k$, so the TDOA between $m_i(t)$ and $m_j(t)$ is obtained as

$$\hat{\tau}_{ij} = \arg \max_k r_{m_i m_j}^{\text{GCC}}(k) \quad (10)$$

where $k \in [-\tau_{\max}, \tau_{\max}]$, and the $\tau_{\max}$ is the maximum possible delay of microphone array.

The frequency-domain weighting function $v(f)$ is calculated straightforwardly using the norm of $\Phi_{m_i m_j}(f)$, because the TDOA information is conveyed in the phase rather than the amplitude of the cross-spectrum [15]. We choose frequency-domain weighting function as

$$v(f) = \frac{1}{|\Phi_{m_i m_j}(f)|}. \quad (11)$$

The amplitude has been normalized since it is not related with TDOA, the phase information which is used for calculating TDOA is left. Thus the GCC can be calculated efficiently.

2) *Steered Response Power*: In our system, since the noise signal is highly uncorrelated, the accuracy of TDOA we obtained from GCC is low. Therefore, the estimation accuracy of the GCC phase transform (PHAT) algorithm is not acceptable, as shown in Section IV. Therefore, in this subsection, we use the SRP [13], [14] algorithm to improve the estimation accuracy.

The signal picked up by the i th microphone is denoted as $m_i(t)$, the SRP of a finite signal frame is defined as [13]

$$P_n(\mathbf{x}) = \int_{nT}^{(n+1)T} \left| \sum_{i=1}^M \omega_i m_i(t - \tau(\mathbf{x}, i)) \right|^2 dt \quad (12)$$

where $\mathbf{x}$ is the 3-D spatial vector of noise source, ω_i is the weight of signal $m_i(t)$, and $\tau(\mathbf{x}, i)$ is the time delay from the noise source directly to the i th microphone.

SRP can be calculated by summing the generalized cross-correlations of all possible microphone pairs of the microphone array [13]. Based on Parseval's theorem, the total energy contained in a waveform $m_i(t)$ summed across all time t is equal to the total energy of the waveform's Fourier Transform, $M(f)$ summed across all of its frequency components f [14]. We transfer (12) into frequency domain as follows [13],

$$\begin{aligned} P_n(\mathbf{x}) &= \sum_{k=1}^M \sum_{l=1}^M \int_{-\infty}^{\infty} W_k(\omega) W_l^*(\omega) M_k(\omega) M_l^*(\omega) \\ &\quad \times e^{j\omega(\tau(\mathbf{x}, l) - \tau(\mathbf{x}, k))} d\omega \end{aligned} \quad (13)$$

where $W_k(\omega)$ and $W_l(\omega)$ are frequency-dependent weights, and $\tau(\mathbf{x}, l) - \tau(\mathbf{x}, k)$ is the TDOA for microphones k and l .

The maximum of $P_n(\mathbf{x})$ will be obtained at the noise source position $\hat{\mathbf{x}}$;

$$\hat{\mathbf{x}} = \arg \max_{\mathbf{x}} P_n(\mathbf{x})$$

$P_n(\mathbf{x})$ is a symmetric matrix with fixed energy terms on the diagonal [8], therefore, we only consider $P'_n(\mathbf{x})$ which

contains the lower triangular part of $P_n(\mathbf{x})$ and changes with $\mathbf{x}$. and changes with $\mathbf{x}$ needs to be concerned as

$$P'_n(\mathbf{x}) = \sum_{k=1}^M \sum_{l=k+1}^M \int_{-\infty}^{\infty} W_k(\omega) W_l^*(\omega) M_k(\omega) M_l^*(\omega) \times e^{j\omega(\tau(\mathbf{x},l) - \tau(\mathbf{x},k))} d\omega. \quad (14)$$

The TDOA information is conveyed in the phase instead of the amplitude of the cross-spectrum [11]. Performing phase transform (PHAT), frequency-dependent weights choose inverse of the magnitude of the cross-spectrum as

$$\psi_{kl}(\omega) = W_k(\omega) W_l^*(\omega) = \frac{1}{|M_k(\omega) M_l^*(\omega)|} \quad (15)$$

and $\tau(\mathbf{x}, l) - \tau(\mathbf{x}, k)$ is the TDOA for microphones k and l .

Searching in a restricted 3-D spatial area for maximum $P'_n(\mathbf{x})$, one will get the estimated location of the noise source. The computational complexity is significantly high for searching for the maximum P'_n in the whole space when the physical space is large. In order to reduce the computational complexity, a suboptimal stochastic region contraction (SRC) searching algorithm is utilized [13].

The basic idea of the SRC algorithm is, given an initial rectangular search volume containing the desired number of global optima and gradually, in an iterative process, decrease the search volume until a sufficiently small sub-volume is reached in which the optimal $P'_n(\mathbf{x})$ is trapped [13].

Define i as the iteration counter, N as the number of random points need to be evaluated in original search volume V_{room} , P ($P \ll N$) as the number of optimal points in volume V_i , V_{min} as the sufficiently small sub-volume in which the optimal $P'_n(\mathbf{x})$ is trapped, and C_{total} as the maximum number of $P'_n(\mathbf{x})$ evaluation allowed in sub volume V_i . The SRC algorithm is illustrated in Fig. 4.

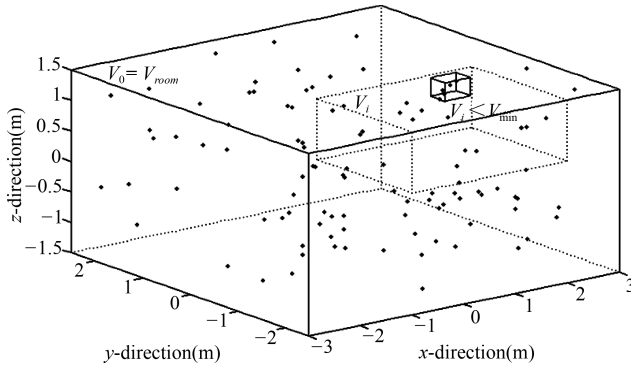


Fig. 4. 3-D SRC search region example.

The SRC search algorithm can be implemented by following steps:

- 1) For the initial iteration $i = 0$, in original volume $V_0 = V_{\text{room}}$, randomly choose N points.
- 2) Calculate $P'_n(\mathbf{x})$ for N points.
- 3) Sort N points' $P'_n(\mathbf{x})$ values, and keep the biggest P points.
- 4) Contract the search region to a smaller volume with boundary $B_i = [(x_{\min}, y_{\min}, z_{\min}), (x_{\max}, y_{\max}, z_{\max})]$ which is determined by the locations of the biggest P points, and compute new region volume V_i .

5) If $V_i < V_{\text{min}}$ stop search and find the optimum location $\hat{\mathbf{x}}$ as: $\hat{\mathbf{x}} = \text{argmax}_{\mathbf{x}} P'_n(\mathbf{x})$

6) Else, calculate the mean value μ_i of $P'_n(\mathbf{x})$ for those P points, save m good points at which $P'_n(\mathbf{x}) > \mu_i$.

7) Continue searching for other $n = P - m$ points in the same region B_i , till sufficient n points have been found or the evaluation times for i th iteration $c_i < C_{\text{total}}$

8) $i = i + 1$, go to Step 4.

3) *Reference Microphone Selection and ANC Algorithm:*

In this paper, we simply choose the microphone close to the noise source as the reference microphone expressed as

$$\mathbf{r} = \arg \min_{\|S - m_i\|} \mathbf{r}_i.$$

Then, the FXLMS algorithm can be utilized to reduce the noise level, the secondary sources are driven by the adaptive filters output signals,

$$y_k(n) = \mathbf{r}_i^T(n) \mathbf{A}_{k,i}(n) \quad (16)$$

where $\mathbf{A}_{k,i}$ is the adaptive filter matrix from i th reference microphone to k th secondary source.

The error signal vector measured by the error microphones is

$$\begin{aligned} \mathbf{e}(n) &= \mathbf{d}(n) + \mathbf{y}'(n) \\ &= \mathbf{d}(n) + \mathbf{S}(n) * [\mathbf{r}^T(n) \mathbf{A}(n)] \end{aligned} \quad (17)$$

where $\mathbf{d}(n)$ is the primary noise vector and $\mathbf{y}'(n)$ is the canceling signal vector at the error sensors.

$$\mathbf{A}(n+1) = \mathbf{A}(n) - \mu \mathbf{r}'(n) \mathbf{e}(n) \quad (18)$$

where μ is the step size,

$$\begin{aligned} \mathbf{r}'(n) &= [\mathbf{S}(n) * \mathbf{r}^T(n)]^T \\ &= \begin{bmatrix} \hat{s}_{11}(n) & \hat{s}_{12}(n) & \cdots & \hat{s}_{1K}(n) \\ \hat{s}_{21}(n) & \hat{s}_{22}(n) & \cdots & \hat{s}_{2K}(n) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{s}_{M1}(n) & \hat{s}_{M2}(n) & \cdots & \hat{s}_{MK}(n) \end{bmatrix} * \begin{bmatrix} \mathbf{0} \\ \vdots \\ \mathbf{r}(n) \\ \vdots \\ \mathbf{0} \end{bmatrix}^T \end{aligned} \quad (19)$$

and $\mathbf{S}(n)$ is the secondary paths matrix among the secondary loudspeakers and error microphones.

IV. CASE STUDY AND EXPERIMENT RESULTS

In this paper, an ANC application for an infant incubator is used as an example of the proposed integrated system. In this system, the microphone array consists of four omnidirectional microphones and is placed at the four corners outside of incubator, as shown in Fig. 5. Recorded neonatal intensive care unit (NICU) noise is used as the noise source is played by a loudspeaker. The TASCAM HS-P82 multi-track recorder is used to record the 4-channel noise signals from the 4 microphones, sampling frequency is 48 kHz, which is down sampled to 6.4 kHz.

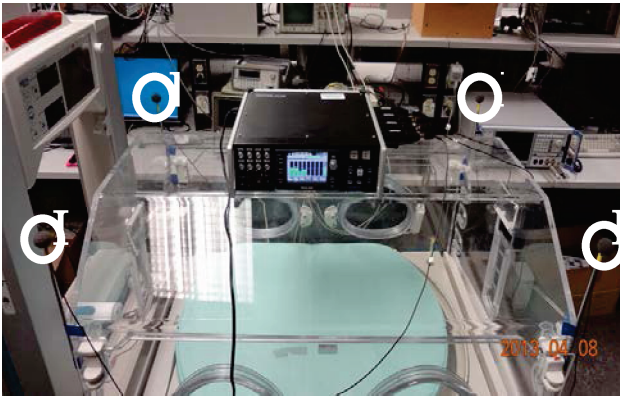


Fig. 5. The experimental set up of noise source direction detection.

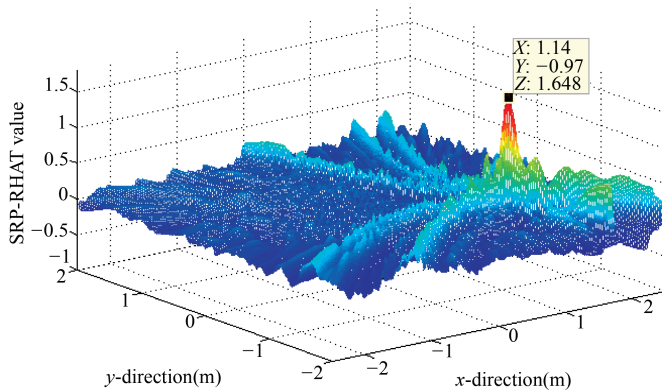


Fig. 6. Searching result (2-D) for maximum.

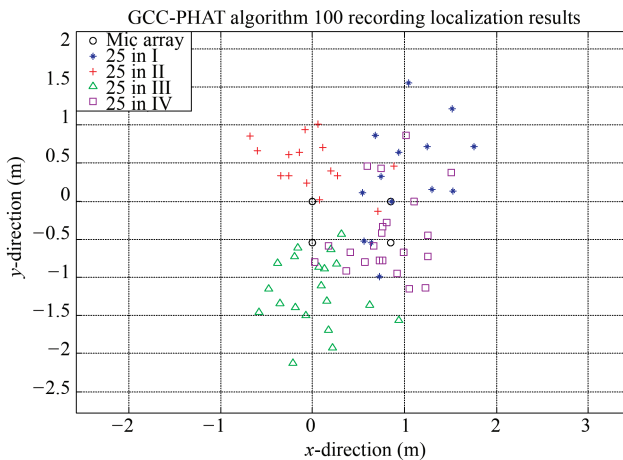


Fig. 7. Estimation of 100 noise source positions by using GCC-PHAT method.

The position of microphone m_2 is assigned as the origin of the coordinate system, and the microphone array locates at $[(0.85, 0, 0); (0, 0, 0); (0, -0.55, 0); (0.85, -0.55, 0)]$. In every quadrant, we randomly change noise source location 25 times and record 4-channel synchronous signals. These recorded signals are processed using GCC-PHAT and SRP-PHAT algorithms.

Fig. 6 shows 2-D searching result for maximum $P'_n(\mathbf{x})$. It is shown that the SRP achieves the highest value at the noise source position. The location of peak SRP's value is the position of the noise source.

In each quadrant, the estimated position for unknown noise

sources are plotted in 2-D plane using different color marks as shown in Fig. 7. We find that 16 points are misestimated. For example, some green points fall in quadrant IV and some estimated locations of the noise sources are inside the range of the microphone array.

The detailed localization estimation results of the GCC-PHAT algorithm are tabulated in Table I. In the computer simulation, 29 of 100 recordings have no real solutions which are indicated as N/A in the table, and 55 out of 71 estimated noise source points are in the right quadrant as highlighted in shaded cells of Table I. The correct rate of noise source localization using GCC algorithm is 55 % and it is not acceptable.

TABLE I
NOISE SOURCE LOCALIZATION BY USING GCC METHOD

		True noise source in			
		Q _I	Q _{II}	Q _{III}	Q _{IV}
Estimated locations near to	m_1	11	2	0	5
	m_2	0	13	0	0
	m_3	0	0	18	4
	m_4	3	0	2	13
	N/A	11	10	5	3

For the SRP algorithm, the estimated results of the same total 100 positions are shown in the 2-D plane of the microphone array in Fig. 8. It is well noticed that the quantity of the wrong estimation for unknown noise sources is reduced. For example, only 5 points fall in other quadrants.

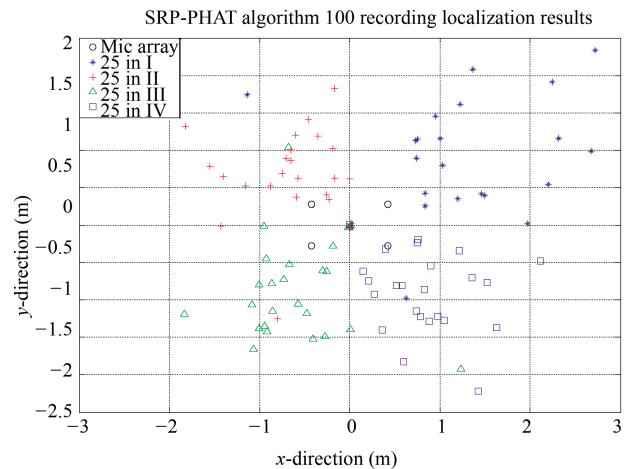


Fig. 8. Estimation of 100 noise source positions by using SRP-PHAT method.

The performance of the SRP algorithm is summarized in Table II. There are 10 estimated locations inside the microphones array area, so these results are invalid and indicated as N/A. There are 5 estimated noise source locations in the wrong quadrant, thus the correct rate of noise source localization using SRP algorithm is 85 %.

Based on the results obtained from SRP algorithm, we selected the microphone close to the noise source to act as reference microphone. The 1X2X2 feedforward ANC system with two error sensors and two secondary loudspeakers is used. We use the recorded NICU noise as the primary noise.

TABLE II
NOISE SOURCE LOCALIZATION BY USING SRP METHOD

		True noise source in			
		Q _I	Q _{II}	Q _{III}	Q _{IV}
Estimated locations near to	m ₁	20	0	0	0
	m ₂	1	21	1	0
	m ₃	0	1	21	0
	m ₄	1	0	1	23
	N/A	3	3	2	2

Table III presents the experimental results of the microphone- array-integrated ANC system's performance when the reference microphone is chosen properly and improperly. When the wrong reference microphone is chosen, the causality of the ANC system is invalid and the performance is degraded. When the ANC system is causal, the performance is improved by around 5 dB.

TABLE III
NOISE REDUCTION LEVELS FOR DIFFERENT REFERENCE SELECTION (THE BOLD ONES ARE CAUSAL ANC SYSTEMS)

Noise source position	Reference microphone	Left error microphone (dB)	Right error microphone (dB)
Q _I	m ₁	9.8	5.7
	m ₃	4.1	2.2
Q _{III}	m ₁	4.5	7.1
	m ₃	9.6	12.2

V. CONCLUSION

We proposed to develop the microphone array integrated ANC system. The performance of a feedforward ANC system is analyzed when the causality condition is violated. A microphone array with 4 microphones is used to pick up the noise signal simultaneously; the GCCT algorithm and SRP algorithm are used to estimate the noise source location and direction for selecting the proper reference sensor. The simulation and experiment results show that the proposed technique can locate the noise sources with high accuracy. The SRC search algorithm is utilized to reduce the computational complexity of the SPR algorithm. The FXLMS algorithm was conducted using the properly selected reference microphone. Our future work includes: multiple noise source detection and reference microphone signals processing for feed forward ANC systems, and conduct a case study for the moving noise source problem with real NICU.

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